

UrbanHeat AI

Al Khor area study, Qatar

Theme: Urban Expansion, Land Use Change & Heat Risk

Satellite indicators for urban-planning site visits

Team Al Zubarah, Qatar

UAE Space Academy hackathon, October 2026

The planning problem

Surface heating and limited greenery can help identify places that warrant a closer look.

Municipal planners need traceable evidence to organise field investigation before proposing interventions.

Pedestrians and outdoor users are the intended beneficiaries. Their exposure and needs have not been measured.

The prototype supports investigation; it does not establish heat-health risk.

Planning use case

A planner needs a shortlist of places to inspect for shade or planting.

The map shows observed surface temperature, green signal and mapped buildings for each cell.

A site visit must establish pedestrian activity, existing shade, ownership and feasibility.

No user interview, measured time saving or cooling benefit is claimed.

Open satellite data and mapped context

Sentinel-2: green-pixel indicators from 5, 15 and 30 September 2026.

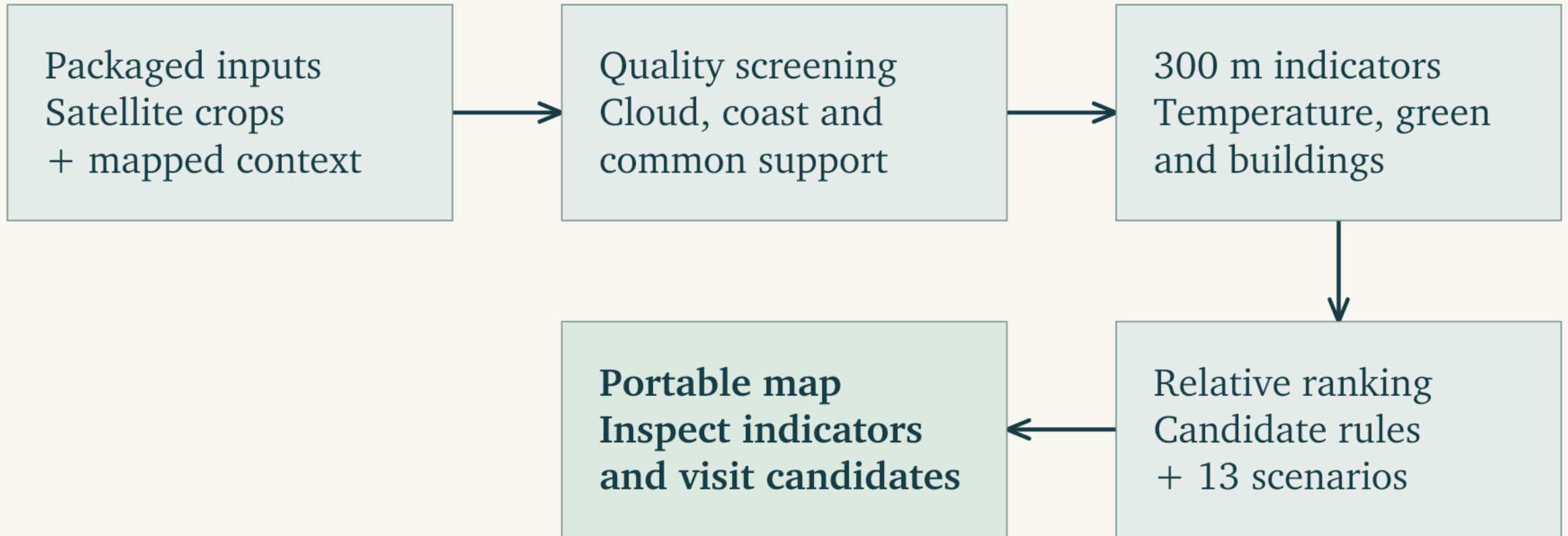
Landsat: surface temperature from 14 and 30 September. The 6 September scene fails coverage QA.

WorldCover 2021: historical land-cover context. OpenStreetMap: contributed building footprints downloaded 5 October.

No hyperspectral, 813 or proprietary VHR data are used.

Thermal native detail is about 100 m, delivered on a 30 m grid. We report 300 m cells.

From source crops to an inspectable shortlist



Deterministic processing; upstream WorldCover uses ML. Our own segmentation model is future work.

Al Khor study window and shortlist

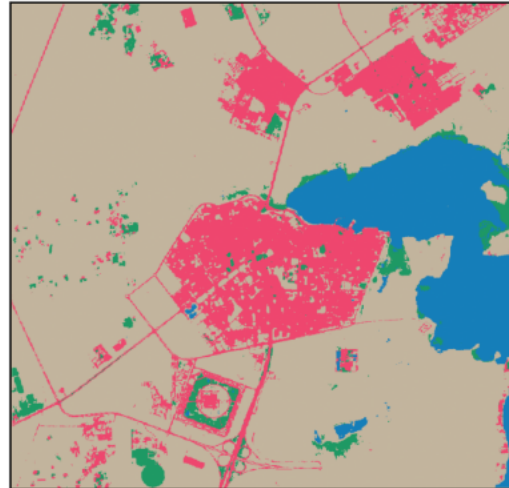
Al Khor · land-cover baseline and planning shortlist

Mapped building footprints · OSM



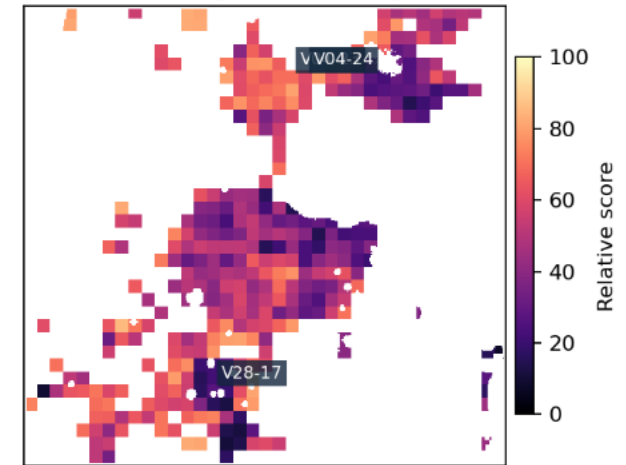
UTM east (m)

Historical land cover · WorldCover 2021



UTM east (m)

Relative investigation score · pilot only



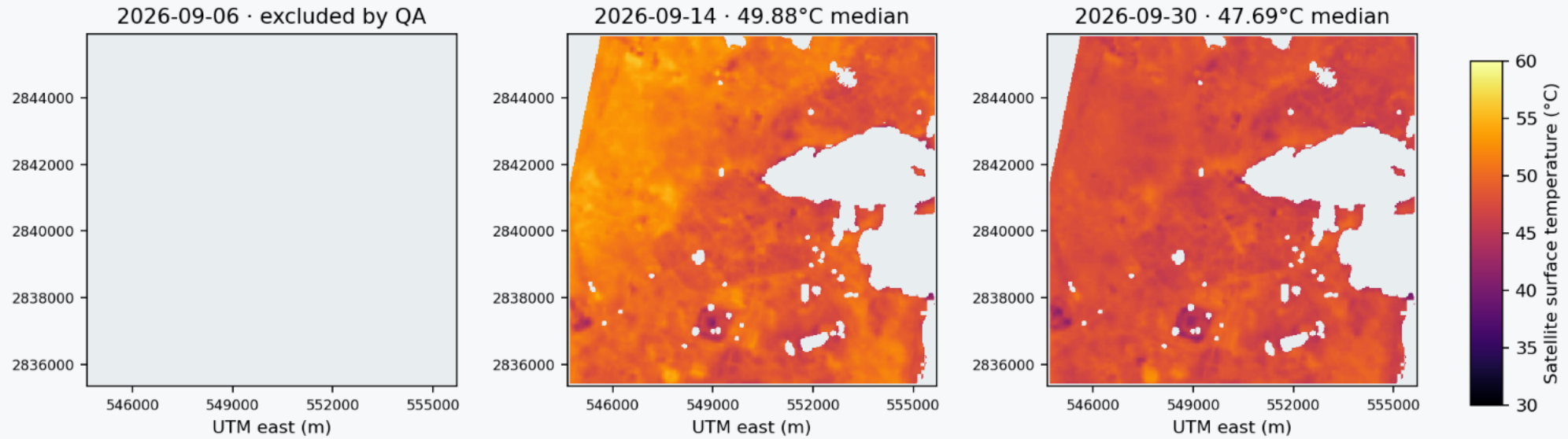
UTM east (m)

WorldCover: blue water, pink built-up, green vegetation, beige other · OSM footprints are contributed and may be incomplete
Shortlist requires site checks for pedestrian use, shade, ownership and planting feasibility. No health-risk or cooling benefit inferred.

1,111 observed cells and 373 screened urban candidates. Analyst-defined window, approximately 116 km².

Matched surface-temperature observations

Surface temperature · 3 dates checked · 2 pass the filters



Morning Landsat observations · 100 m thermal detail on a 30 m grid · Same colour scale
Water/coastal/uncertain pixels excluded; date-to-date differences are not a long-term warming trend or proof of vegetation cooling.

94.4379 km² of common inland sample footprints across the broad study window.

Leading candidates change with the settings

Candidate	Surface °C*	Green %*	Buildings %	Rank range
V04-23	50.4	0.0	2.3	1-14
V28-17	50.0	0.0	0.1	1-23
V04-24	50.2	0.1	2.9	2-34

Score: 50% temperature percentile, 30% inverse green percentile, 20% mapped-building percentile.

Thirteen sensitivity settings show no uniquely verified best intervention site.

*Means of date-level cell summaries. Rank ranges are scenario ranges, not confidence intervals.

Reproducibility and reference checks

The package includes 35 hashed input files, source metadata and pinned Python 3.12 dependencies.

Clean GitHub execution passed through a normal Jupyter kernel: five cells, zero errors and all 35 input hashes verified.

Grass reference agreement: 24/25. Building green-rule agreement: 0/30. Coastal points retained in thermal support: 0/30.

Contributed reference labels have conflicts. Human confirmation and independent thermal validation remain pending.

Potential impact and incubation next steps

Proposed benefit: help planners inspect the evidence behind a shortlist before arranging field visits. No time saving or cooling benefit has been measured.

Next pilot: confirm boundaries and reference labels, visit candidate sites and collect planner feedback.

Incubation: assess suitable VHR data, implement and validate segmentation, then test shade and paving indicators.

The current PoC does not measure urban expansion, health risk, causal cooling or long-term warming.

Team responsibilities for final review

Nasser Alhuri: team lead, integration, submission and pitch.

Abdulrahman Almohannadi: data and method review.

Mohammed Almarri: map and candidate-site review.

Majed Alkuwari: reproducibility check and demo review.

Ali Alkubaisi: presentation and package quality review.

Assigned review ownership, not claims of completed contributions. Code, analysis and drafts were developed with AI assistance.

Data attribution and sources

Contains modified Copernicus Sentinel data (2026). Optical source: Element 84 Earth Search.

Landsat imagery courtesy of USGS. Collection 2 Level-2 surface temperature, doi:10.5066/P9OGBGM6.

ESA WorldCover 2021 v200, Zanaga et al. (2022). CC BY 4.0, doi:10.5281/zenodo.7254220.

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Full scene IDs, exact provenance, processing rules and limitations accompany the package. Original code: MIT.